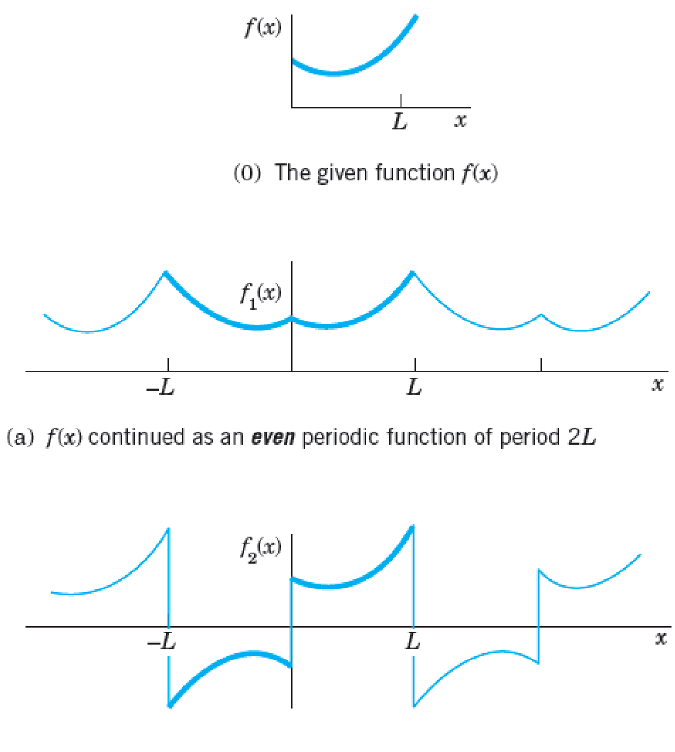
3. Half-Range Expansions

Half-range expansions are Fourier series. The idea is simple and useful. Figure 270 explains it. We want to represent $f(x)$ in Fig. 270.0 by a Fourier series, where $f(x)$ may be the shape of a distorted violin string or the temperature in a metal bar of length L, for example. (Corresponding problems will be discussed in Chap. 12.) Now comes the idea.

We could extend $f(x)$ as a function of period L and develop the extended function into a Fourier series. But this series would, in general, contain *both* cosine *and* sine terms. We can do better and get simpler series. Indeed, for our given f we can calculate Fourier coefficients from (6*) or from (6**). And we have a choice and can take what seems more practical. If we use (6*), we get (5*). This is the **even periodic extension** f_1 of f in Fig. 270a. If we choose (6**) instead, we get (5**), the **odd periodic extension** f_2 of f in Fig. 270b.

Both extensions have period $2L$. This motivates the name **half-range expansions**: f is given (and of physical interest) only on half the range, that is, on half the interval of periodicity of length $2L$.

Let us illustrate these ideas with an example that we shall also need in Chap. 12.



(b) $f(x)$ continued as an ***odd*** periodic function of period $2L$

Fig. 270. Even and odd extensions of period $2L$

EXAMPLE 6

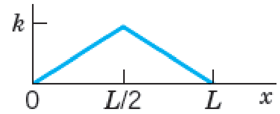


Fig. 271. The given function in Example 6

"Triangle" and Its Half-Range Expansions

Find the two half-range expansions of the function (Fig. 271)

$$f(x) = \begin{cases} \dfrac{2k}{L}x & \text{if} \quad 0 < x < \dfrac{L}{2} \\[2ex] \dfrac{2k}{L}(L-x) & \text{if} \quad \dfrac{L}{2} < x < L. \end{cases}$$

Solution. (a) ***Even periodic extension.*** From (6*) we obtain

$$a_0 = \frac{1}{L}\left[\frac{2k}{L}\int_0^{L/2} x\,dx + \frac{2k}{L}\int_{L/2}^{L}(L-x)\,dx\right] = \frac{k}{2},$$

$$a_n = \frac{2}{L}\left[\frac{2k}{L}\int_L^{L/2} x\cos\frac{n\pi}{L}x\,dx + \frac{2k}{L}\int_{L/2}^{L}(L-x)\cos\frac{n\pi}{L}x\,dx\right].$$

We consider a_n. For the first integral we obtain by integration by parts

$$\int_0^{L/2} x\cos\frac{n\pi}{L}x\,dx = \frac{Lx}{n\pi}\sin\frac{n\pi}{L}x\,\Bigg|_0^{L/2} - \frac{L}{n\pi}\int_0^{L/2}\sin\frac{n\pi}{L}x\,dx$$

$$= \frac{L^2}{2n\pi}\sin\frac{n\pi}{2} + \frac{L^2}{n^2\pi^2}\left(\cos\frac{n\pi}{2} - 1\right).$$

Similarly, for the second integral we obtain

$$\int_{L/2}^{L}(L-x)\cos\frac{n\pi}{L}x\,dx = \frac{L}{n\pi}(L-x)\sin\frac{n\pi}{L}x\,\Bigg|_{L/2}^{L} + \frac{L}{n\pi}\int_{L/2}^{L}\sin\frac{n\pi}{L}x\,dx$$

$$= \left(0 - \frac{L}{n\pi}\left(L - \frac{L}{2}\right)\sin\frac{n\pi}{2}\right) - \frac{L^2}{n^2\pi^2}\left(\cos n\pi - \cos\frac{n\pi}{2}\right).$$

We insert these two results into the formula for a_n. The sine terms cancel and so does a factor L^2. This gives

$$a_n = \frac{4k}{n^2\pi^2}\left(2\cos\frac{n\pi}{2} - \cos n\pi - 1\right).$$

Thus,

$$a_2 = -16k/(2^2\pi^2), \qquad a_6 = -16k/(6^2\pi^2), \qquad a_{10} = -16k/(10^2\pi^2), \cdots$$

and $a_n = 0$ if $n \neq 2, 6, 10, 14, \cdots$. Hence the first half-range expansion of $f(x)$ is (Fig. 272a)

$$f(x) = \frac{k}{2} - \frac{16k}{\pi^2}\left(\frac{1}{2^2}\cos\frac{2\pi}{L}x + \frac{1}{6^2}\cos\frac{6\pi}{L}x + \cdots\right).$$

This Fourier cosine series represents the even periodic extension of the given function $f(x)$, of period $2L$.

(b) ***Odd periodic extension.*** Similarly, from (6**) we obtain

This Fourier cosine series represents the even periodic extension of the given function $f(x)$, of period $2L$.

(b) ***Odd periodic extension.*** Similarly, from (6**) we obtain

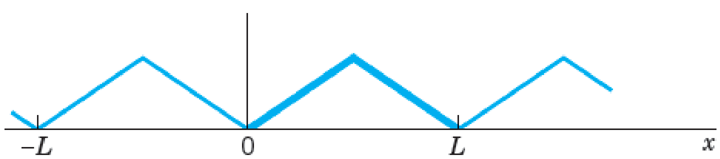
(5) $$b_n = \frac{8k}{n^2\pi^2}\sin\frac{n\pi}{2}.$$

Hence the other half-range expansion of $f(x)$ is (Fig. 272b)

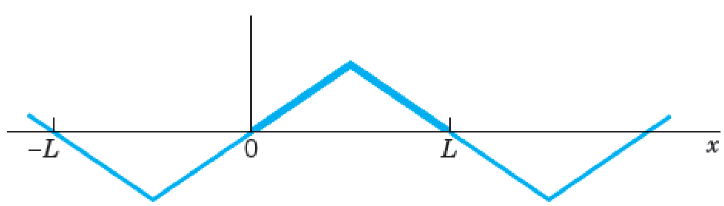
$$f(x) = \frac{8k}{\pi^2}\left(\frac{1}{1^2}\sin\frac{\pi}{L}x - \frac{1}{3^2}\sin\frac{3\pi}{L}x + \frac{1}{5^2}\sin\frac{5\pi}{L}x - + \cdots\right).$$

The series represents the odd periodic extension of $f(x)$, of period $2L$.

Basic applications of these results will be shown in Secs. 12.3 and 12.5.



(a) Even extension



(b) Odd extension

Fig. 272. Periodic extensions of $f(x)$ in Example 6

Complex Fourier Series

In this section we show that the Fourier series

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx) \quad \text{—} (1)$$

can be written in the form of complex, which sometimes simplifies calculations (See Ex #1 on page 498). This complex form can be obtained because in complex, the exponential function e^{it} and $\cos t$ and $\sin t$ are related by the basic Euler formula

$$e^{it} = \cos t + i\sin t \qquad \text{thus } e^{-it} = \cos t - i\sin t \rightarrow (2)$$

Conversely, by adding and subtracting these two formulas, we obtain

$$\left.\begin{aligned} \cos t &= \frac{e^{it} + e^{-it}}{2} \rightarrow (a) \\ \sin t &= \frac{e^{it} - e^{-it}}{2i} \text{ — } (b) \end{aligned}\right] (3)$$

From (3) $\frac{1}{i} = -i$ in $\sin t$ and setting $t = nx$ in both formulas, we get

$$a_n \cos nx + b_n \sin nx = \frac{a_n}{2}\left(e^{inx} + e^{-inx}\right) + \frac{b_n}{2i}\left(e^{inx} - e^{-inx}\right)$$

$$= \frac{1}{2}(a_n - ib_n)e^{inx} + \frac{1}{2}(a_n + ib_n)e^{-inx}$$

We insert this into (1) writing $a_0 = c_0$

$\frac{1}{2}(a_n - ib_n) = c_n$ and

$\frac{1}{2}(a_n + ib_n) = k_n$

We get from (1)

$$f(x) = c_0 + \sum_{n=1}^{\infty}\left(c_n e^{inx} + k_n e^{-inx}\right) \longrightarrow (4)$$

The coefficients $c_1, c_2 \ldots$ and $k_1, k_2, \ldots$ are obtained from (6b), (6c) in Sec 11.1 and then (2) above with $t = nx$,

(5)

$$c_n = \frac{1}{2}(a_n - ib_n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)[\cos nx - i\sin nx]\,dx$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)\, e^{-inx}\, dx.$$

$$k_n = \frac{1}{2}(a_n + ib_n) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x)[\cos nx + i\sin nx]\,dx$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{inx}\, dx.$$

Finally, we can combine eq (5) into a single formula by the trick of writing $k_n = c_{-n}$, Then (4) and (5) and $c_0 = a_0$ in (6a) of Sec 11.1 give (Summation from $-\infty$!)

$$\left.\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty} c_n e^{inx}, \\ c_n &= \frac{1}{2\pi}\int_{-\pi}^{\pi} f(x) e^{-inx}\,dx, \quad n = 0, \pm 1, \pm 2, \ldots \end{aligned}\right\} \rightarrow (6)$$

This is the so-called complex form of the Fourier Series or, more briefly, the complex Fourier series, of $f(x)$. Then the c_n are called the complex Fourier co-efficients of $f(x)$.

For a function of period $2L$ our reasoning gives the complex Fourier Series.

$$(7)\left\{\begin{aligned} f(x) &= \sum_{n=-\infty}^{\infty} c_n e^{\frac{in\pi x}{L}}, \\ c_n &= \frac{1}{2L}\int_{-L}^{L} f(x) e^{-in\pi x/L}\,dx, \quad n = 0, \pm 1, \pm 2, \ldots \end{aligned}\right.$$

Example No 1 :- Complex Fourier Series

Find the complex Fourier Series of $f(x) = e^{x}$ if $-\pi < x < \pi$ and $f(x+2\pi) = f(x)$ and obtain from it the usual Fourier Series.

Sol:-

Since $\sin n\pi = 0$ for every integer n, we have

$$e^{\pm in\pi} = \cos n\pi \pm i \overset{0}{\cancel{\sin n\pi}} = \cos n\pi = (-1)^{n} \quad \text{---} \; (\star)$$

With this we obtain from (6) by integration

$$C_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{x} e^{-inx}\,dx = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{(1-in)x}\,dx$$

$$\Rightarrow \frac{1}{2\pi} \cdot \left. \frac{e^{(1-in)x}}{(1-in)} \right|_{-\pi}^{\pi}$$

$$\Rightarrow \frac{1}{2\pi} \cdot \frac{1}{(1-in)} \left[e^{(1-in)\pi} - e^{-(1-in)\pi} \right]$$

$$\Rightarrow \frac{1}{2\pi(1-in)} \left[e^{\pi} e^{-in\pi} - e^{-\pi} \cdot e^{in\pi} \right]$$

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By using (*)

AS

$$C_n = \frac{1}{2\pi}\cdot\frac{1}{(1-in)}\left[e^{\pi} - e^{-\pi}\right](-1)^n$$

On the right,

$$\frac{1}{(1-in)} = \frac{1+in}{(1-in)(1+in)} = \frac{1+in}{1+n^2}$$

and $\quad e^{\pi} - e^{-\pi} = 2\sinh\pi$

Hence the Complex Fourier Series is

$$e^x = \frac{\sinh\pi}{\pi}\sum_{n=-\infty}^{\infty}(-1)^n\frac{(1+in)}{1+n^2}e^{inx} \longrightarrow (8)$$

$-\pi < x < \pi$

From this let us derive the real Fourier Series. Using (2) with $\boxed{t = nx}$ and $i^2 = -1$ we have. in (8)

$$(1+in)e^{inx} = (1+in)\left[\cos nx + i\sin nx\right]$$

$$= (\cos nx - n\sin nx) + i(n\cos nx + \sin nx)$$

Now (8) also has a corresponding term with $-n$ instead of n. Since $\cos(-nx) = \cos nx$

$\sin(-nx) = -\sin nx$

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we obtain in this term

$$(1-in)e^{-inx} = (1-in)[\cos nx - i\sin nx]$$

$$= (\cos nx - n\sin nx) - i(n\cos nx + \sin nx)$$

If we add these two expressions, the imaginary parts cancel, Hence their sum is

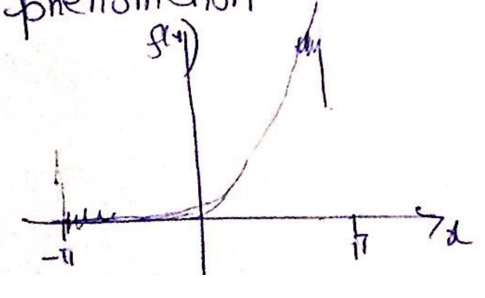
$$2[\cos nx - n\sin nx], \qquad n=1,2,\cdots$$

For $\boxed{n=0}$ we get 1 (not 2) because there is only one term. Hence the real Fourier series is

(9)

$$e^x = \frac{2\sinh\pi}{\pi}\left[\frac{1}{2} - \frac{1}{1+1^2}(\cos x - \sin x) + \frac{1}{1+2^2}(\cos 2x - 2\sin 2x) - \cdots\right]$$

In the Fig. 270 the poor approximation near the jumps at $\pm\pi$ is a case of the Gibbs phenomenon



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Recommended Ex: 11.3

Complex Fourier Series

Exercise.

Find the Complex Fourier Series of the following function. (Show the detail of your work).

QNo1 $f(x) = \begin{cases} -1 & \text{if } -\pi < x < 0 \\ 1 & \text{if } 0 < x < \pi \end{cases}$

Ans: $-\dfrac{2i}{\pi} \displaystyle\sum_{n=-\infty}^{\infty} \frac{1}{(2n+1)} e^{(2n+1)ix}$.

QNo2 $f(x) = x \quad (-\pi < x < \pi)$

Ans. $i \displaystyle\sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{(-1)^n}{n} e^{inx}$.

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Q No 3 $f(x) = x^2$ $(-\pi < x < \pi)$

Ans. $$\frac{\pi^2}{3} + 2 \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{(-1)^n}{n^2} e^{inx}$$

Q No 4 $f(x) = x$ $(0 < x < 2\pi)$

Ans: $$\pi + i \sum_{\substack{n=-\infty \\ n \neq 0}}^{\infty} \frac{1}{n} e^{inx}.$$